

## **MATH 313: PRINCIPLE OF OPERATIONS RESEARCH**

### **Course content**

**Introduction; Linear Programming; Methods of solving LPP; Simplex method; Convex Analysis in  $E^n$ ; Concepts in Economics e.g. Demand, Supply, Elasticity, Duality e.t.c.; Graph Theory; Transportation and Assignment Problems.**

### **References**

- 1) Operations Research Applications and Algorithms by Wayne L. Winstone**
- 2) Operations Research Techniques by S. French, R. Halley, L. C. Thomas and D. T. White**
- 3) Operations Research by S. S. Cohen**

### **Introduction**

Decision making in today's social and business environment has become a complex task. The uncertainty of the future and the nature of competition greatly increase the difficulty of managerial decision making. Operations research as one of the quantitative aid to decision making offers a decision maker a method of evaluating every possible alternative (act or course of acting) by using various techniques to know the potential outcome.

### **History of Operations Research**

Operations research came into existence as a discipline during the World War II when there was a critical need to manage scarce resources. However, a particular model can be traced back to much earlier times. The term Operations research was coined as a result of research on military operations during this war. Since the war involved strategic and tactical problems, groups of individuals who collectively considered specialized in Mathematics, Economics, Statistics, Engineering, Behavioral and Physical Sciences were formed as special units within the armed forces.

### **Definition 1: Operations research**

- 1) Operations research may be described as a scientific approach to decision making that involves the operations of the organizational system ( *F. S. Hiller and G. J. Lieberman*)

- 2) Operations research is essentially a collection of mathematical techniques and tools which in conjunction with a system approach are applied to solve practical decision problems of an economic and engineering nature (*Deallanbach and George 1978*)
- 3) Operations research is characterized by the use of scientific knowledge through interdisciplinary team effort for the purpose to determining the best utilization of limited resources (*H. A. Taha 1976*)
- 4) Operations research is the art of winning wars without actually fighting them (*Clark*)
- 5) Operations research is the art of finding bad answers to problems to which otherwise worse answers are given (*T. L. Saaty*)

Other names that are given to operations research include: *Operations analysis, System analysis, Decision analysis, Management science, Decision science* e.t.c. each of these emphasizes the quantitative approach to the analysis and solution of management problems

### Assignment

- 1) Discuss the significance of operations research in your area of speciality/profession
- 2) Discuss the limitations of operations research

## LINEAR PROGRAMMING (LP)

### Definition: Linear programming

Linear programming is a class of mathematical models concerned with the efficient allocation of limited resources to known activities with the objective of meeting a defined goal such as maximizing of profit or minimizing of cost.

In LP, we attempt to maximize or minimize (optimize) a linear function of decision variables. This function is known as *objective function*. The values of the decision variables must satisfy a set of constraints. Each constraint must be a linear equation or a linear inequality

**Definition: Linear function:-** A function  $f(x_1, x_2, \dots, x_n)$  of  $X_1, X_2, \dots, X_n$  is a linear function if

for some constant  $c_1, c_2, \dots, c_n$ ,  $f(x_1, x_2, \dots, x_n)$  is given by  $c_1x_1 + c_2x_2 + \dots + c_nx_n$ .

**Definition: Linear equation:-** a linear equation consists of a linear function  $f(x_1, x_2, \dots, x_n)$  a constant b and the equality symbol =

**Definition: Linear inequality:-** a linear inequality consists of a linear function  $f(x_1, x_2, \dots, x_n)$  a constant b and the inequality symbol ( $<, \leq, \geq$  or  $>$ )

### Characteristics of Linear Programming Models (LPM)

LP models have certain characteristics in common. The knowledge of these characteristics enables us to recognize problems that are amenable to a solution using linear programming models. Moreover, same knowledge is necessary for one to be able to formulate a LP model.

These characteristics are grouped into two

- (i) Components
- (ii) Assumptions

The components relate to the structure of the model while assumptions reveal the conditions under which the model is valid.

### Components of LP Models

LP models have four components namely;

- i) *Objective function:* - it is a criterion by which the values all the decision variables are evaluated. In LP the objective is either maximization or minimization
- ii) *Decision variables:* -these are variables that control the value of the objective function through choices in their levels. They represent unknown quantities to be determined
- iii) *Constraints:* -these are restrictions or limits that influence the ability of a decision maker to select values of the decision variables in a LP model, e.g. availability of resources such as raw materials, labour, time, space e.t.c. Constraints arise from government policies, forecasts, customer orders.
- iv) *Parameters:* -these are fixed values that specify the impact that one unit of each decision variable has on the objective function and on any constraint it pertains.

## **Assumptions of LP Models**

There are four basic assumptions in LP models namely;

- i) *Linearity assumption:* -it is assumed that each decision variable has a linear impact on the objective function and on each constraint in which it appears.
- ii) *Divisibility assumption:* -it is assumed that integer and non-integer values are acceptable as potential solutions to a LP model.
- iii) *Certainty assumption:* -this assumption involves two aspects of LPM's. first it is assumed that the parameters are assumed to be known and fixed. Secondly, it is assumed that all relevant constraints have been identified and represented in the model.
- iv) *Non-negativity assumption:* -it is assumed that only positive values and zero are allowed as potential solutions to a LP model. Negative values of the decision variables are non- realistic and therefore cannot be considered as any potential solution.